THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 1 & 2: **Exploring Substances: Acidic, Basic, and Neutral, Electricity: Circuits and Their Components** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Red litmus turns ___ in a basic solution. 1

(a) red **(b)** blue **(c)** yellow **(d)** pink

2. Common salt solution is: 1

(a) acidic **(b)** basic **(c)** corrosive **(d)** neutral

3. Citrus fruits such as lemon contain: 1

(a) formic acid **(b)** lactic acid **(c)** citric acid **(d)** acetic acid

4. The colour of turmeric paper in an acidic solution: 1

(a) stays yellow **(b)** turns green **(c)** turns red **(d)** turns blue

5. Indigestion is caused by excess: 1

(a) acid in the stomach **(b)** salt in food
(c) base in the stomach **(d)** water in the stomach

6. The thin wire inside a bulb that glows is the: 1

(a) terminal **(b)** filament **(c)** switch **(d)** electrode

7. A device used to open or close a circuit is a: 1

(a) filament **(b)** switch **(c)** terminal **(d)** fuse only

8. Materials through which current passes easily are: 1

(a) solvents **(b)** indicators **(c)** conductors **(d)** insulators

9. Why are electrical wires covered with plastic? 1

(a) Plastic is an insulator and protects from shock
(b) It looks good
(c) Plastic is a conductor
(d) It makes them heavy

10. The current from a cell flows from: 1

(a) no direction
(b) negative to positive only
(c) both ends
(d) positive terminal to negative terminal through the circuit

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 15–25 words each.

11. How do farmers treat acidic soil? 2

12. What is an olfactory indicator? Give an example. 2

13. What are the terminals of a cell? 2

14. Why is it dangerous to touch electric devices with wet hands? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. State the safety rules while working with chemicals in the lab. 3

16. Why does a bulb in a torch glow brighter with fresh cells? 3

17. Why does lemon juice taste sour while soap solution tastes bitter? 3

OR

What is the role of acid rain in damaging monuments? 3

18. State two uses of conductors and two of insulators. 3

OR

A learner tests a steel spoon, a plastic ruler and a pencil lead. Which make the bulb glow? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) What is an olfactory indicator? 5
(ii) Describe the activity with onion strips.
(iii) Observations with acid and base.

OR

Case study: A gardener finds hydrangea flowers change colour with soil. 5
(i) What property is shown?
(ii) What is the soil type for blue?
(iii) How can colour be altered?
(iv) Name a natural indicator.

20. (i) Why is a closed loop needed for current flow? 5
(ii) What happens when a wire breaks?
(iii) Name a device that breaks the circuit on purpose.

OR

Case study: A battery of two cells powers a toy. 5
(i) What is a battery?
(ii) How is it connected?
(iii) What if one cell is reversed?
(iv) What is the effect on brightness?

END OF QUESTION PAPER — Total: 40 Marks

NEET MIND